Lluís Saló-Salgado, PhD

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Research Interests

I conduct research at the intersection of geology, flow in porous media, and geomechanics to mitigate geohazards and address outstanding scientific challenges related to energy development in complex subsurface systems.

In particular, my work focuses on developing physics-based and data-driven computational tools—anchored with field and experimental observations—to elucidate the interplay between geologic discontinuities, fluid migration, and seismicity. I seek to enable sustainable development of engineered geosystems for the energy transition.

Keywords: Geosystems engineering, geoenergy, fault hydrogeology, multiphysics reservoir simulation

Education

2024 **Massachusetts Institute of Technology**, Cambridge, MA, USA
PhD in Geotechnical and Geoenvironmental Engineering
Faculty advisor: Ruben Juanes
Thesis: *Numerical Modeling of Geologic Carbon Dioxide Storage in Faulted Siliciclastic Settings*

2016 **Universitat Politècnica de Catalunya**, Barcelona, Spain
MS in Geotechnical and Earthquake Engineering

2014 **Universitat de Barcelona**, Barcelona, Spain
BS in Geology

Professional Appointments

01/2026– **Assistant Professor**
Barbara McDonald Endowed Professorship in Energy Transition
Dept. of Earth, Environmental, and Planetary Sciences • The University of Tennessee, Knoxville

2023–2025 **Postdoctoral Fellow** • Dept. of Earth and Planetary Sciences • Harvard University
Postdoctoral Associate • Dept. of Civil and Environmental Engineering • MIT
Faculty mentors: John Shaw (Harvard) and Ruben Juanes (MIT)

Fellowships and Honors

2021 "la Caixa" Foundation Postgraduate Fellowship • "la Caixa" Foundation
2020 MathWorks Engineering Fellowship • The MathWorks & SoE at MIT
2017 Eni-MIT Energy Fellowship • MIT Energy Initiative
2017 Michael F. Parlamis (1962) and Goldberg-Zoino (1954) Combined Fellowship • MIT CEE
2016 Highest GPA in the MS cohort in Geotechnical and Earthquake Engineering (9.7/10) • UPC

Research

Invited Talks

2026-12	(Sched.) AGU Fall Meeting, San Francisco, CA	American Geophysical Union
2026-10	(Sched.) Virginia Tech	Geosciences, seminar
2026-06	Limits and Promise of Geonergies, Palma, Spain	IMEDEA-CSIC-UIUC
2026-05	Universitetet i Bergen	Porous Media Group, seminar
2026-04	The University of Tennessee, Knoxville	Civil and Environmental Engineering, seminar
2026-03	The University of Tennessee, Knoxville	Bredesen Center, seminar
2026-02	The University of Texas at Austin	Petroleum and Geosystems Engineering, graduate seminar
2025-11	Harvard University	Earth and Planetary Sciences, colloquium
2024-12	Perkins-Rosen Conference, Houston, TX	GCS-SEPM
2024-07	The University of Texas at Austin (virtual)	Gulf Coast Carbon Center, PREDICT workshop
2024-06	The University of Texas at Austin (virtual)	Gulf Coast Carbon Center, PREDICT workshop
2024-01	Massachusetts Institute of Technology (virtual)	MIT Energy Initiative, guest lecture
2023-11	Harvard University	Earth and Planetary Sciences, BiSEPPS seminar
2023-04	Massachusetts Institute of Technology	Earth Resources Laboratory, FISH seminar
2022-09	Universitat Politècnica de Catalunya	Civil and Environmental Engineering, GHS seminar
2022-09	ICGC, Barcelona, Spain	Energy Transition & Geohazards colloquium
2022-05	SINTEF Digital, Oslo, Norway	Computational Geosciences, seminar
2021-10	Universitetet i Bergen	Porous Media Group, seminar

Grants

2026–2027 *"CO_2 Migration along Faults in CCS Projects"*, co-PI (with R. Juanes). Funded by ExxonMobil Technology and Engineering Company.

2026–2027 *"Site Assessment Methodology for CCS Suitability in the Central Valley, CA"*, co-I (PI: J. H. Shaw), subaward through Harvard. Funded by the California Department of Conservation.

Supervision

Color indicates the supervisee's primary institution: UTK, Harvard, MIT.

Postdoctoral Researchers

2026– Rubén Vidal Montes Supervised research: Computational modeling of fault hydraulic properties for CO_2 storage

2026– L. T. Ozgur Yildirim Supervised research: Experimental modeling and characterization of clay-smeared fault–fluid interactions

Undergraduate Students

2025– Angela Liu (co-advised with Hannah Lu and Ruben Juanes). MIT Undergraduate Research Opportunities Program: Enhancing Modeling of Subsurface Fluid Flow in Fractured Rocks through Machine Learning

2025 Penelope Salmon (co-advised with John H. Shaw). Supervised research: Pore pressure evolution during hydrogen storage in a depleted hydrocarbon field

2024–2025 Chloe Fair (co-advised with Natasha Toghramadjian and John H. Shaw). B.S. Thesis: "Utilizing Abandoned Oil and Natural Gas Wells in the LA Basin for Carbon Sequestration and Hydrogen Storage"

2024–2025 Ben Littlejohn (co-advised with Andreas Plesch and John H. Shaw). Harvard Program for Research in Science and Engineering: Evaluation of Potential Underground Hydrogen Storage Sites in the Los Angeles Basin
- Ben published an article in *IJHE* based on this work

2024–2025 Julia Mansfield (co-advised with Andreas Plesch and John H. Shaw). B.S. Thesis: "A Comparative Assessment of the Capacity for CO_2 Storage in Depleted Hydrocarbon Reservoirs of the Los Angeles Basin, California, Using Multiphase Flow Simulations"
- Julia was awarded Harvard FAS's *Hoopes Prize* for her honors thesis

2024 Riddhi Bhagwat (co-advised with Hannah Lu and Ruben Juanes). MIT Undergraduate Research Opportunities Program: Enhancing Subsurface Fluid Flow and Solute Transport Modeling through Machine Learning: A Study of Fractured Rocks Using FracNet

2023–2024 Joyce Qu (co-advised with Hannah Lu and Ruben Juanes). MIT Undergraduate Research Opportunities Program: Sensitivity Analysis of Lithological Controls and Surrogate Modeling for Fault Permeability in Siliciclastic Sediments

2022–2024 Runako Gentles (co-advised with Hannah Lu and Ruben Juanes). MIT Undergraduate Research Opportunities Program: Clustering of carbon dioxide injection-induced earthquakes using feature-based representation of time series data

Memberships

2026– International Society for Porous Media (InterPore)

2026– Society of Petroleum Engineers (SPE)

2025– Society for Industrial and Applied Mathematics (SIAM)

2020– Geological Society of America (GSA)

2018– American Geophysical Union (AGU)

Teaching

Instructor

UTK

EEPS 5XX: Simulation of Subsurface Systems (graduate)

Fall 2026

EEPS 690: Seminar in Earth & Environmental Sciences (graduate)

Spring 2026

Harvard

(Guest Instructor) EPS 272: Topics in Structural Geology & Earthquake Science (graduate)

I gave lectures and led discussion sessions (Fall 2025)

MIT

(Lab Instructor) 1.000: Introduction to Computer Programming and Numerical Methods for Engineering Applications (undergraduate)

Language: Python. I gave lectures, hosted weekly office hours, and graded assignments (Fall 2024)

Teaching Assistant

MIT

TA in 1.723: Computational Methods for Flow in Porous Media (graduate)

Hosted weekly office hours and graded all assignments (Fall 2020, Fall 2022)

UPC TA in Geology (undergraduate)

Supervised petrology and geological mapping labs (Spring 2016)

Pedagogical Training

2026	Early Career Geoscience Faculty Workshop Program, NAGT
2026	Adult Mental Health First Aid Certificate, UTK
2024	Research Mentoring Certificate, MIT
2023	Kaufman Teaching Certificate Program, MIT

Service

Conferences

2026 (Scheduled) Primary session convener, *"Coupled Fault Processes in Carbon Sequestration, Hydrogen Storage, and Geothermal Energy"*, AGU Fall Meeting, San Francisco, CA

(Scheduled) Co-organizer, *"Fault Zones and Fluid Flow, from Outcrops to Algorithms"*, GSA Connects, Denver, CO

(Scheduled) Co-organizer, *"Graph Neural Networks and Digital Twins for Geological Storage of CO_2"*, SIAM Conference on Mathematics of Planet Earth, Cleveland, OH

Member of the Technical Committee, *HydroML Symposium*, hosted by UT Austin, TX

Co-organizer, *"MS03: Flow, Transport and Mechanics in Fractured Porous Media"*, InterPore, Nantes, France

2025 Primary session convener, *"Interplay between Fault Architecture and Coupled Processes in Carbon Sequestration, Hydrogen Storage, and Geothermal Energy"*, AGU Fall Meeting, New Orleans, LA

2024 Primary session convener, *"Predictive modeling and risk assessment in subsurface geoenergy technologies"*, AGU Fall Meeting, Washington, DC

Discussion leader, *"Communicating your research to support solutions to the climate and energy transition"*, Gordon Research Seminar in Flow and Transport in Permeable Media, Newry, ME

2023 Primary session convener, *"Fault zone fluid migration and induced seismicity potential in geoenergy technologies"*, AGU Fall Meeting, San Francisco, CA

Committees

2026–	Member of the Energy Science and Engineering Admissions Committee, Bredesen Center, UTK
2026–	Member of the Graduate Admissions Program Committee, EEPS, UTK
2020–2022	Treasurer, Spain@MIT
2018–2020	Graduate student committee member, MIT Civil and Environmental Engineering

Outreach/Career Mentoring

2026	Speaker, Knox Climate Watch: "Managing Fault Hazards in Subsurface Energy Systems"
2024	Graduate Application Assistance Program mentor, MIT Civil and Environmental Engineering
2023	K-12 student mentor, MIT Museum: "Computer science applications in the 21st century"
2023	Speaker, Harvard Museum of Natural History: "Earthquakes and their impacts in society"
2021	Panelist, MIT Civil and Environmental Engineering: "Communication in graduate school"

Reviewer

Journals	Energy & Fuels, Water Resources Research, Geophysical Research Letters, Journal of Hydrology, SPE Journal, iScience, Journal of Geophysical Research: Solid Earth, Earth and Space Science, International Journal of Greenhouse Gas Control, Transport in Porous Media, Engineering Geology, Scientific Reports, Environmental Science: Processes & Impacts, Natural Hazards & Earth System Sciences
Conferences	HydroML (2026), American Rock Mechanics Association (2023, 2024), American Geophysical Union OSPA program (2023)
Books	EGS the Future Energy Road Ahead (CRC Press, 2023)

Publications

* indicates students I advised.

In Review

2. H. Lu, **Lluís Saló-Salgado**, Y.-T. Chou, E. Haghighat, and R. Juanes. Learning and inferring multiphase flow dynamics in porous media using scientific machine learning: Application to the "FluidFlower" CO_2 injection experiment, 2026a. Submitted for publication. Preprint available at 10.48550/arXiv.2606.05448

1. **Lluís Saló-Salgado**, J. A. Silva, A. Plesch, F. D. Wolfe, J. H. Shaw, and R. Juanes. Stress state, subsidence, and faulting in the Wilmington Oil Field, California: a multiphase flow–geomechanics modeling assessment (1936–2020), 2026. Submitted for publication. Preprint available at 10.48550/arXiv.2605.16711

Published

12. H. Lu, **Lluís Saló-Salgado**, and R. Juanes. Lithological controls on the permeability of geologic faults: surrogate modeling and sensitivity analysis. *Stochastic Environmental Research and Risk Assessment*, 40(8): 186, 2026b. `10.1007/s00477-026-03305-z`

11. B. Littlejohn*, **Lluís Saló-Salgado**, A. Plesch, and J. H. Shaw. Integrated geologic and socio-economic evaluation of potential underground hydrogen storage sites in the Los Angeles Basin. *International Journal of Hydrogen Energy*, 202:152964, 2026. `10.1016/j.ijhydene.2025.152964`

10. H. Lu, **Lluís Saló-Salgado**, Y. M. Marzouk, and R. Juanes. Uncertainty quantification of fluid leakage and fault instability in geologic CO_2 storage. *Water Resources Research*, 61(10):e2024WR039275, 2025. `10.1029/2024WR039275`

9. **Lluís Saló-Salgado**, J. A. Silva, L. Lun, C. M. Rogers, J. S. Davis, and R. Juanes. Assessing CO_2 migration within faults during megatonne-scale geologic carbon dioxide storage in offshore Texas. *Water Resources Research*, 61(5):e2024WR037059, 2025a. `10.1029/2024WR037059`

8. **Lluís Saló-Salgado**, O. Møyner, K.-A. Lie, and R. Juanes. Three-dimensional simulation of geologic carbon dioxide storage using `MRST`. *Advances in Geo-Energy Research*, 14(1):34–48, 2024b. `10.46690/ager.2024.10.06`. [SI: MRST]

7. M. Haugen, **Lluís Saló-Salgado**, K. Eikehaug, B. Benali, J. W. Both, E. Storvik, O. Folkvord, R. Juanes, J. M. Nordbotten, and M. Fernø. Physical variability in meter-scale laboratory CO_2 injections in faulted geometries. *Transport in Porous Media*, 151:1169–1197, 2024. `10.1007/s11242-023-02047-8`. [SI: FluidFlower]

6. **Lluís Saló-Salgado**, M. Haugen, K. Eikehaug, M. Fernø, J. M. Nordbotten, and R. Juanes. Direct comparison of numerical simulations and experiments of CO_2 injection and migration in geologic media: Value of local data and forecasting capability. *Transport in Porous Media*, 151:1199–1240, 2024a. `10.1007/s11242-023-01972-y`. [SI: FluidFlower]

5. J. A. Silva, **Lluís Saló-Salgado**, J. Patterson, G. R. Dasari, and R. Juanes. Assessing the viability of CO_2 storage in offshore formations of the Gulf of Mexico at a scale relevant for climate-change mitigation. *International Journal of Greenhouse Gas Control*, 126(6):103884, 2023. `10.1016/j.ijggc.2023.103884`

4. **Lluís Saló-Salgado**, J. S. Davis, and R. Juanes. Fault permeability from stochastic modeling of clay smears. *Geology*, 51(1):91–95, 2023. `10.1130/G50739.1`

3. J. A. Gili, R. Ruiz-Carulla, G. Matas, J. Moya, A. Prades, J. Corominas, N. Lantada, M.-A. Núñez Andrés, F. Buill, C. Puig, J. Martínez-Bofill, **Lluís Saló**, and O. Mavrouli. Rockfalls: analysis of the block fragmentation through field experiments. *Landslides*, 2022. `10.1007/s10346-021-01837-9`

2. **Lluís Saló**, J. Corominas, N. Lantada, G. Matas, A. Prades, and R. Ruiz-Carulla. Seismic energy analysis as generated by impact and fragmentation of single-block experimental rockfalls. *Journal of Geophysical Research: Earth Surface*, 123(6):1450–1478, 2018. `10.1029/2017JF004374`.
   † Featured in journal cover

1. **Lluís Saló**, T. Frontera, X. Goula, L. G. Pujades, and A. Ledesma. Earthquake static stress transfer in the 2013 Gulf of Valencia (Spain) seismic sequence. *Solid Earth*, 8:857–857, 2017. `10.5194/se-8-857-2017`

Other Writing

- **Lluís Saló-Salgado**, A. Acocella, I. Arzuaga García, S. El Mousadik, and A. Zvinavashe. Managing up: how to communicate effectively with your PhD adviser. *Nature*, 2021. `10.1038/d41586-021-03703-z`

Recent Conference Talks

- H. Lu, **Lluís Saló-Salgado**, Y. M. Marzouk, and R. Juanes. Uncertainty quantification of fluid leakage and fault instability in geologic CO_2 storage. In *InterPore 18th Annual Meeting, May 19–22, Nantes, FRA*. International Society for Porous Media, 2026c

- **Lluís Saló-Salgado**, J. A. Silva, A. Plesch, J. H. Shaw, and R. Juanes. Coupled flow and geomechanics modeling of ground deformation and fault stability at the Wilmington field, CA, 1936-2020. In *AGU Fall Meeting, December 15–19, New Orleans, LA*. American Geophysical Union, 2025b

- **Lluís Saló-Salgado**, J. A. Silva, A. Plesch, J. H. Shaw, and R. Juanes. Coupled flow and geomechanics modeling of ground deformation and fault stability at the Wilmington field, CA, 1936-2020. In *SIAM Geosciences, October 14–17, Baton Rouge, LA*. Society for Industrial and Applied Mathematics, 2025c